Received: April 02, 2023 | Accepted: April 29, 2023 | Published: May 06, 2023

Crossref PEERCITE PUBLISHERS Open Access

REVIEW

Peercite Journal of Infectious and Rare Disease

SILVER NANOPARTICLES: A VERSATILE WEAPON AGAINST INFECTIONS AND CANCER

Ardhendu Kumar Mandal

Central Instrumentation Division, CSIR-Indian Institute of Chemical Biology, India

*Corresponding Author : Ardhendu Kumar Mandal

Email :ardhendu_mandal@yahoo.co.in

Citation: Mandal AK (2023) Silver Nanoparticles: a versatile weapon against infections and cancer. Peercite J Infect Rare Dis. 1(2):4005-4009.

Copyright: © 2023 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://peercite.org).

Abstract :

Despite advances in medicine and pharmacy, chemotherapy still poses serious problems in delivering drugs to specific target sites for many life-threatening infections. Most drugs exhibit significant toxicity, which causes a variety of side effects and affects quality of life. Inadequate therapeutic indices, impaired drug absorption, development of multidrug resistance, and adverse systemic side effects are all associated with the use of conventional fungicides to treat infections. Antimicrobial silver nanoparticles (AgNPs) have been proven to be potent and effective agents against infectious diseases due to their controlled ultrasmall size, large surface area, and enhanced reactivity with active functional structures. After exposure, AgNPs attach to cells and penetrate into cells, producing reactive oxygen species (ROS) that cause oxidative stress, protein dysfunction, altered signaling pathways, and damage membranes and deoxyribonucleic acid (DNA) and cause cell damage. The surface ligand coating of drugs incorporated into AgNPs as drug delivery vehicles allows sustained release of drugs while reducing side effects upon administration into the body. This review mainly focuses on the synthesis, surface functionalization, mechanism of action, biomedical applications, therapeutic efficacy, toxicity, and potential of AgNPs as nanomedicine drug delivery systems for targeted therapy against infectious diseases and cancer. We focus on biodistribution and elimination.

Keywords : Infectious diseases, Silver nanoparticles, Drug delivery system

Introduction :

When microorganisms such as viruses, bacteria, and parasites invade the body, the innate immune system attacks them by using phagocytes, especially macrophages, to search for infections before they become apparent. It acts as a first line of defense against infectious agents. Once these initial defenses have weakened, the lymphocyte-mediated adaptive immune system kicks in and responds specifically to each infectious agent and eradicates it. It also prevents future illnesses by remembering infectious agents [1]. During an infection, the concentration of C-reactive protein (CRP) increases in the blood. This CRP then binds to the C protein in the microbial cell wall, promoting complement fixation and facilitating uptake of the microorganism by phagocytes such as macrophages through chemotactic opsonization.

Disruption of the microbial cell wall within the phagolysosomal compartment of macrophages by proteolytic lysozyme facilitates attack by the complement system on the cell membrane.

Viral infections:

Interferons (IFNs) form a group of proteins that are important in viral infections. When a host cell is infected with a virus, IFN may be produced. Different cell types produce IFN-α or IFN-β during infection, whereas T lymphocytes produced in the thymus release their IFN-γ upon activation by antigen. These IFNs collectively act on uninfected cells to induce an antiviral resistance state [2].

In cell-mediated immunity, antigen-presenting cells present processed antigens to helper T cells, which are central to the generation of an immune response. These phenomena help B cells produced in the bone marrow and fetal liver to produce antibodies and other cells, including natural killer cells, granulocytes, macrophages, cytotoxic T cells, and antibody-dependent cytotoxic cells. It may enhance the anti-infective activity of various effector cells. Many of these effects are mediated by lymphokines and cytokines, particularly macrophages, but both T and B cells can be influenced by suppressor T cells [3].

During disease progression, microorganisms proliferate within the phagolysosomes of host macrophages by evading the proteolytic effects of lysozyme and suppressing the host's protective immune responses. Infectious diseases caused by intracellular or extracellular infections, biofilms, or transmission through medical devices therefore represent a global public health threat, causing millions of deaths each year. Although the use of antibiotics in the 20th century has significantly reduced population morbidity and mortality from infectious microorganisms,

the development of antimicrobial resistance poses a growing public health problem [4]. The development of new antibiotics or the chemical modification of existing antimicrobials can not only solve the problem of microbial resistance, but also require long-term effective metal nanotechnology in medicine against infectious diseases.

Highly toxic in microorganisms:

Silver nanoparticles (AgNPs) in suspension release silver ions that are highly toxic to microorganisms. This bactericidal activity depends on the AgNP structure, dimensions, size, concentration, ionic strength, coating, temperature, and time of dissolution behavior. Their antibacterial effects occur through damage to the microbial membrane by adhesion to the cell surface, followed by functional and structural changes such as gap formation, membrane destabilization, transmembrane and cytoplasmic leakage. Intracellular structures of microorganisms-damage caused by release of free Ag+ ions, subsequent ROS generation, and inactivation by strong macromolecules such as enzymes, proteins, and nucleotides.

Furthermore, these materials with surface coatings can be used as carriers to deliver a wide range of therapeutic components such as drugs, antibodies, and pharmaceuticals in various biomedical applications [5].

The controlled release of biologically active silver from nanosilver can be controlled by various ligand coatings to prevent the release of silver ions and extend the half-life in the systemic circulation of passive targeting. Furthermore, surface functionalization of these AgNPs can be achieved by modifying various ligands such as sugars, proteins, peptides, and genetic materials, and their therapeutic efficacy requires further research to achieve active targeting. Overall, this review provides an effective guide to reduce toxicity, improve delivery, improve solubility and bioavailability, and provide better formulations for synergistic effects against pathogenic diseases and cancer. We focus on ligand-coated drug-loaded AgNPs as oral therapeutic tools.

Preparation of nanodrug conjugates:

Nanodrug conjugates incorporated into AgNPs are chemically maintained by reduction of metal salt precursors with stirring for 2 h. spin the homogeneous slurry at 4000 rpm for 25 min to obtain pellets. The pellet will be lyophilized and used for further studies [6].

Preparation of Ligand-Coated Silver-Nano-Drug Complex:

Chitosan-coated silver-nano-drug complex is prepared by an ionotropic gelation process. Mix the silver nanodrug complex with 100 mg of chitosan and 1.5 mg of acetic acid. Then add sodium triphosphate solution dropwise and stir for 2 hours. The resulting slurry is freeze-dried and used for further studies [5].

Citric acid-capped AgNPs are synthesized according to method while adding 5 mM sodium citrate and 25 μM tannic acid to 250 μM AgNO3 solution. The mixture is continuously stirred and refluxed until the solution becomes pale yellow. Citrate AgNPs were purified by ultrafiltration, washed, and stored at 4°C for future use. Other types of ligands such as mercaptopropionic acid, mercaptohexanoic acid, mercaptopropione sulfonic acid, polyvinylpyrrolidone, polyethyleneimine, mono-, di-, or polyethylene glycols and sugars are also used to coat AgNPs.

In green synthesis, non-toxic lignin is used as both capping agent and reducing agent for the formation of silver lignin nanoparticles (AgLNPs) [2]. Briefly, lignin solutions (0.01, 0.1, and 1% w/v) were prepared in Milli-Q water and the pH was adjusted to 5.5 with 5 M hydrochloric acid (HCl).

A gLNPs are then prepared by adding 20 mL of 2 mg/mL AgNO3 aqueous solution in Milli-Q water to 30 mL of each lignin solution and running the reaction at 60°C for 3 days with stirring [7].

Characterization of Silver Nanoparticle Composites:

Characterization is performed to determine the surface chemistry, surface area and morphology and property discrepancies of AgNP composites. The formation of AgNP complexes is evaluated by UV-Vis spectroscopy at wavelengths (absorption bands) of approximately 200-800 nm. Size distribution and particle size are evaluated by dynamic light scattering. Morphology is examined using high-resolution transmission electron microscopy. Topography and morphology are observed using scanning electron microscopy [8]. The surface chemistry and the involvement of biomolecules in the synthesized nanoparticles are examined using Fourier transform infrared spectroscopy. The structure of crystalline metal nanoparticles is observed by X-ray diffraction analysis. The elemental composition of the nanoparticles is analyzed using energy dispersive X-ray spectroscopy [4]. We will use Auger electron spectroscopy to analyze the surface oxidation state of silver that composes the hybrid material. The structure and elemental composition of the nanoparticle complexes are analyzed by low-energy ion scattering.

Mechanism of action of silver nanoparticles:

Silver has always been known for its broad spectrum antibacterial effects. The antibacterial potency of AgNPs was increased by modifying the surface dimensions to reduce the size to less than 10 nm. This can also help fight multidrug resistance, where multidrug resistance proteins and P-glycoproteins are responsible for drug efflux. Various studies have been attempted to elucidate the mechanism of action of AgNP against pathogenic microorganisms (figure 1) [9].

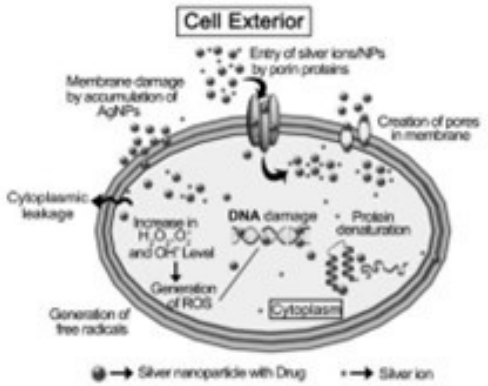

Figure 1: Probable interactive mechanism of actions of silver nanoparticles against microbes.

Under ambient conditions, O2 molecules chemically adsorb onto the 111 planes of the AgNP surface and oxidize the Ag atoms on the surface to form Ag+ ions, while O2 molecules absorb active oxygen species such as O2. The released positive silver ions can bind to negatively charged cell membranes, thereby compromising membrane integrity. Furthermore, AgNPs undergoing endocytosis or pinocytosis can exhibit Trojan horse-like effects by releasing silver ions in the cytoplasm and interacting with intracellular organelles [1,9]. Positively charged Ag+ ions form complexes with nucleic acids and interact with nucleosides. Intracellular Ag+ ions released by AgNPs can bind to thiol groups (-SH) of proteins, glycoproteins, and enzymes on the cell surface, resulting in cell membrane destabilization and loss of respiratory enzymes. Formation of ROS, which causes disruption of mitochondrial ATP synthesis and induces oxidative stress. Ag+ ions can also attach to membrane walls and create pores, through which they can enter intracellular microorganisms and interact with intracellular components and sulfur-containing proteins [10].

Biomedical Applications of Silver Nanoparticles:

Silver nanoparticles and their complexes have been shown to modulate various activities such as antiviral, antibacterial, antiparasitic, antibiofilm, anticancer, antifouling, and drug delivery systems (Figure 2).
Author declares there is no conflict of interest.

Figure 2: Diverse applications of silver nanoparticles.

Aacivities in Drug Delivery Systems:

Drug delivery refers to transporting cargo to a desired location in order to achieve a potential therapeutic effect. A variety of nanoparticle-based formulations have been reported that can be used to target drugs to specific sites for various diseases. Various types of biocompatible and biodegradable conjugated NPs have been reported to be used for drug targeting against inflammation, infection, and cancer. AgNPs combining doxorubicin and alendronate have been used as drug delivery systems to achieve intracellular delivery and improve anticancer therapeutic index. Other studies have shown the use of hybridized Fe3O4-AgNPs as strong magnetic thermo mediators [11,12].

Toxicity of Silver Nanoparticles:

AgNPs cause membrane damage, ROS generation, protein denaturation, mitochondrial dysfunction, DNA damage, glutathione deficiency, lactate dehydrogenase leakage,inhibition of cell proliferation and subsequent stimulation; May be toxic if exposed to cells. Apoptotic and necrotic cell death [13]. However, the extent of its toxicity depends on physicochemical properties such as particle size, shape, surface chemistry, crystallinity, surface functionalization including capping agents, ligands, dispersion rate, concentration, surface charge and environmental pH. Masu and macromolecules depen. In vivo studies showed that smaller (10 nm) AgNPs exhibited higher levels of acute cytotoxicity in histopathological changes in the form of cellular inflammation, congestion, focal and isolated liver necrosis, and splenic congestion [14].

Conclusion:

Traditional antimicrobial therapy with antibiotics and other drugs poses problems regarding acquired drug resistance by infectious microorganisms in several diseases. While the focus has been on the development of new antibiotics and the chemical modification of existing drugs to solve this problem, metal nanoparticles are unique due to their ultra-small size, high surface area-to-volume ratio, and unique physical chemistry. Properties, it has emerged as a new potential antibacterial agent. Through interactions with microorganisms, such as cellular uptake and aggregation, leading to toxicity and death of microorganisms. Ligand-dependent Ag+ release by the drug can provide potent synergistic antibacterial activity not only to the drug but also to his AgNPs due to the short carbon chain and weak binding with oxygen. Therefore, optimization of surface ligands such as coordinating atoms, carbon chain length, and end groups is of great importance to prepare nanoparticles for commercial applications against infectious diseases.

Furthermore, to minimize side toxicity and achieve higher biological efficiency, AgNPs can be surface-functionalized with specific ligands and coatings to target specific sites for sustained charge release. It can be used for effective biomedical applications as drug delivery systems. However, to achieve maximum in vivo efficacy of ligand-bound charge-loaded AgNPs, optimal formulation, dosage, biocompatibility, pharmacokinetics, biodistribution, biodegradation, excretion, toxicity and especially the route of administration require further Long-term research is essential. Administer orally and intravenously before going to the clinic for nanomedicine translation.

Conflict of interest:

Author declares there is no conflict of interest.

References:

1. Cohen ML (2000) Changing patterns of infectious disease. Nature. 406:762-767.

2. Ferrari M (2005) Cancer nanotechnology: opportunities and challenges. Nat Rev Cancer. 5:161-171.

3. Bouromeester H, Poortman J, Peters RJ et al. (2011) Characterization of translocation of silver nanoparticles and effects on whole-genome gene expression using an in vitro intestinal epithelium coculture model. ACS Nano. 5:4091-4103.

4. Loeschner K, Hadrup N, Quortrup K et al. (2011) Distribution of silver in rats following 28 days of repeated oral exposure of silver nanoparticles or silver acetate. Part Fibre Toxicol. 8:18.

5. Liu J, Sonstine DA, Shervani S et al. (2010) Controlled release of biologically active silver from nanosilver surfaces. ACS Nano. 4:6903-6913.

6. Liu J, Hurt RH (2010) Ion release kinetics and particle persistence in aqueous nano-silver colloids. Environ Sci Technol. 44:2169-2175.

7. Kittler S, Greulich C, Diendorf J et al. (2010) Toxicity of silver nanoparticles increases during storage because of slow dissolution under release of silver ions. Chem Mater. 22:4548-4554.

8. Akter M, Sikder MT, Rahman MM et al. (2018) Asystematic review on silver nanoparticles-induced cytotoxicity: Physicochemical properties and perspectives. J Adv Res. 9:1-16.

9.Zheng K, Setyawati MI, Leong DT et al. (2018) Antimicrobial silver nanomaterials. Coord Chem Rev. 357:1-17.

10.Duran N, Duran M, deJesus MB et al. (2016) Silver nanoparticles: A new view on mechanistic aspects on antimicrobial activity. Nanomed Nanotechnol Biol Med. 12:789-799.

11.Badawy AME, Luxton TP, Silve RG et al. (2010) Impact of environmental conditions (pH, ionic strength, and electrolyte type) on the surface charge and aggregation of silver nanoparticles suspensions. Environ Sci Technol. 44:1260-1266.

12.Levard C, Hotze EM, Lowry GV et al. (2012) Environmental transformations of silver nanoparticles: impact on stability and toxicity. Environ Sci Technol. 46:6900-6914.

13. Almatroudi A (2020) Silver nanoparticles: Synthesis, characterization and biomedical applications. Open Life Sci. 15:819-839.

14.Marambio-Jones C, Hoek EMV (2010) A review of the antibacterial effects of silver nanomaterials and potential implications for human health and the environment. J Nano Res. 12:1531-1551.